Southern Cross University

The Endocrine System

The adrenal cortex: The stress axis

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COMMONWEALTH OF AUSTRALIA

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Specific objectives

- Name and explain the actions of the two types of steroidal hormones released from the adrenal **cortex** (glucocorticoids and mineralocorticoids – the androgens are not as important so concentrate on the first two).
- Using hypoglycaemia as the stressor explain why the 'hypothalamus > anterior pituitary > adrenal cortex' axis is considered the stress axis (in other words, how does this axis help out in hypoglycaemia?)
- Using dehydration or hypovolemic shock as the stressor explain why the 'hypothalamus > anterior pituitary > adrenal cortex' axis is considered the stress axis (How does the adrenal gland help in times of dehydration?) Drugs to know: Spironolactone and Fludrocortisone
- Detail the physiological actions of the glucocorticoids. Give specific examples of the application of glucocorticoids in pharmacology, using specific drugs as examples (*Methylprednisolone, Prednisolone and Dexamethasone).* List ten specific adverse effects of glucocorticoids.

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Medications of the Endocrine System: The adrenal cortex:

Lesson 1

Overview of the stress axis and mineralocorticoids

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'Corticosteroids'

Steroidal hormones secreted by the adrenal cortex

They include:

Glucocorticoids
Save glucose

Mineralocorticoids
Save salt

NOTE: In clinic, you may refer to 'corticosteroids' as being synonymous with the pharmaceutical glucocorticoids like prednisone

Note: Androgens like testosterone are secreted by the adrenal cortex as well but they are never referred to as corticosteroids – they are sex hormones

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Overview

What is meant when a person says "I have had to be on steroids"?

Corticosteroids are steroidal hormones secreted by the adrenal cortex. They include:

- Glucocorticoids and Mineralocorticoids

- Anabolic steroids are testosterone and its 'analogues' (similar in structure)
  - A person would not say they are on anabolic steroids

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Overview

What is meant when a person says "I have had to be on steroids"?

Corticosteroids are steroidal hormones secreted by the adrenal cortex. They include:

- Glucocorticoids and Mineralocorticoids

The glucocorticoids include cortisol, the major stress hormone

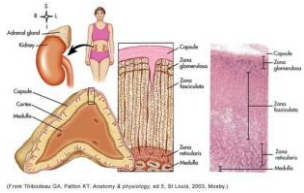
- 'Glucocorticoids' refers to cortisol and similar
- 'Cortisol' is the most abundant glucocorticoid
- Prednisone and prednisolone are medications that act like cortisol
  - Taken to decrease the immune system response (asthma, autoimmune disorders, organ transplant…)

- Anabolic steroids are testosterone and its 'analogues' (similar in structure)
  - A person would not say they are on anabolic steroids

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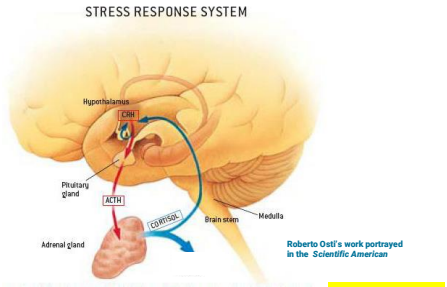
Adrenal Glands

- Adrenal cortex
  - 80% of an adrenal gland's total weight
    - Synthesizes and releases steroid hormones called corticosteroids
- Adrenal medulla
  - Innervated by the sympathetic and parasympathetic nervous systems



(From Thibodeau GA, Patton KT: Anatomy & physiology, ed 5, St Louis, 2003, Mosby.)

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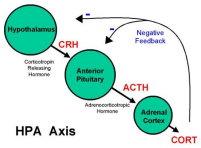


Glucocorticoids, mineral corticoids and noradrenaline are released from the adrenal gland in response to stress

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How the stress axis works

- CRH released from the hypothalamus in response to:
  - Stress
  - Hypoglycemia
  - Decreased circulating levels of glucocorticoids
- Adrenocorticotropin hormone (ACTH) is released from the pituitary when stimulated by CRH, releasing
  - Glucocorticoids,
  - Mineralocorticoids (salt savers)
  - Androgens
- ...From the adrenal cortex



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Medications of the Endocrine System: The adrenal cortex:

Lesson 2

Glucocorticoids

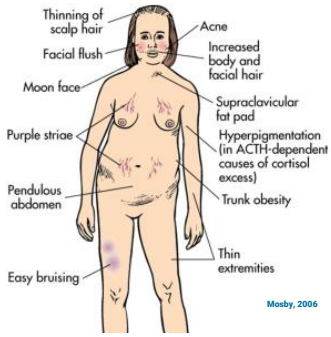
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Effects of glucocorticoids

(Mainly cortisol; a steroidal hormone)

- Cortisol is essential in times of trauma and stress:
  - Affects glucose metabolism
  - Stimulates gluconeogenesis
  - Increases blood glucose
  - Stimulating tissues to use free fatty acids for energy instead of glucose
  - Stimulates protein breakdown
  - Inhibits protein syntheses in all body cells
  - Stimulates hunger
  - Promotes fat buildup in the trunk and facial regions
  - Increases the effects of catecholamines - increased heart rate / blood pressure
  - Inhibition of the immunity and inflammatory
  - Strong effect on emotional stability and mood.

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Medications of the Endocrine System: The adrenal cortex

Lesson 3

Understanding the side effects of corticosteroids

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Alterations in adrenal function

Disorders of the adrenal cortex
- Cushing disease
  - Excessive anterior pituitary secretion of ACTH
- Cushing syndrome
  - Excessive level of cortisol, regardless of cause
  - Iatrogenic Cushing's syndrome can occur with corticosteroid treatment

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Clinical manifestations of Cushing's

- Altered fat metabolism leading to fat pads on the back, moon face, protruding abdomen with thin extremities, stretch marks on breasts, thighs, and abdominal surface.
- Muscle weakness from protein breakdown
- Hypertension as a result of increased catecholamine responsiveness
- Weight gain resulting from strong appetite stimulation. Because of effects on hepatic gluconeogenesis, a reversible form of diabetes mellitus may result.
- **Inhibition of immune and inflammatory reactions**, leading to poor wound healing



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Long term side effects of corticosteroids are the same as Cushing's syndrome

- Altered fat metabolism leading to fat pads on the back, moon face, protruding abdomen with thin extremities, stretch marks on breasts, thighs, and abdominal surface.
- Muscle weakness from protein breakdown
- Hypertension as a result of increased catecholamine responsiveness
- Weight gain resulting from strong appetite stimulation. Because of effects on hepatic gluconeogenesis, a reversible form of diabetes mellitus may result.
- **Inhibition of immune and inflammatory reactions**, leading to poor wound healing



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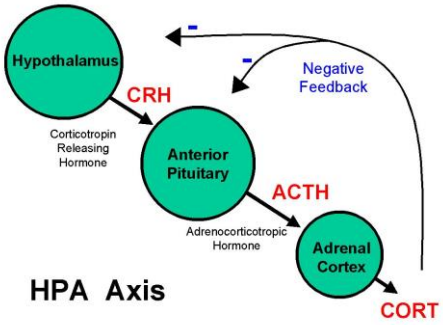
Adrenal dysfunction after long term corticosteroid treatment

- Secondary adrenal insufficiency - hypopituitarism / hypothalamic dysfunction
- ACTH is not released
- Lack of mostly glucocorticoids, but possibly also androgens and mineralocorticoids

This can occur if cortisol treatment is withdrawn suddenly



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Clinical manifestations of secondary adrenal insufficiency

Following sudden withdrawal from corticosteroid treatment, the client may experience:

- Depression
  (because cortisol levels influence mood and emotions)
- Fatigue,
  (related to hypoglycemia, and decreased gluconeogenesis)
- Anorexia, vomiting, diarrhea, and nausea
- Inability to respond to stressful situations
  (perhaps leading to severe hypotension and shock)

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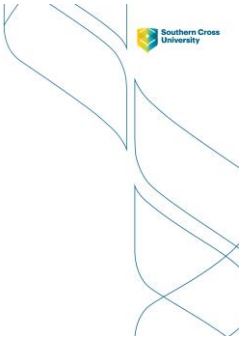
Specific glucocorticoid medications

Prednisolone is a synthetic derivative of cortisol

- Used to treat inflammatory and auto-immune system conditions
- Active metabolite of **prednisone**
- Used with people who cannot metabolise prednisone
  - Liver failure

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Adrenal cortex antagonists



Spironolactone

- Antagonist to the mineralocorticoid aldosterone
- Anti-salt saver (a diuretic)
- (to be discussed at a later time)

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Review: Adrenocortical hormones and their effects

Hormone	Effect
Mineralocorticoids ○ *Aldosterone.* × *Fludrocortisone* ○ *Aldosterone antagonist* × *Spironolactone*	Sodium and water retention
Glucocorticoids ○ *Cortisone, hydrocortisone* × *Methylprednisolone* × *Prednisolone* × *Dexamethasone*	Increase blood glucose Regulate immune/inflammatory response Stimulate protein catabolism

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References

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